

QUESTÃO 1 - timer0

$F_{osc} = 4 \text{ MHz}$ $CM = \frac{F_{osc}}{4} = \frac{4}{4} = 1 \mu s$

Ataudo. Desc. Judo = 80 ms

$\Delta t = CM * \text{Prescaler} * (256 - TMR0)$

$1:128 * (256 - 100) = 19968 \approx 32$

$\Delta t = 1 \mu s * 128 * (256 - 100) = 19968 \cdot 10^{-6} \approx 19,968 \text{ ms}$

$1:256 * (256 - 100) = 19968 \approx 32$

OPTION_REG:

1	1	0	1	0	1	1	0
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Interrupt Edge Rising
Int0 pull-up pin
Ativa Prescaler Prescaler Probação
Enable clock interna (crystal de quartzo)

tempo de incremento = $1 \mu s$

Prescaler = 1:128

TMR0 = 100

OPTION_REG = 11010110

QUESTÃO 2 - timer1

$F_{osc} = 8 \text{ MHz}$ $F_{osc}/4 = \frac{8}{4} = 2 \text{ MHz}$ $CM = \frac{1}{2 \text{ M}} = 0,5 \mu s$

FREQUÊNCIA ↓ tempo

Ataudo Desc. Judo = 50 ms (temporizador)

REGISTADOR TMR1 (16 bits) = TMR1H (8 bits) + TMR1L (8 bits)

$\Delta t = CM * \text{Prescaler} * (65536 - TMR1)$

$65536 - \frac{TMR1}{1:2} = 50000$

T1CON:

-	-	0	1	1	1	0	1
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32 Prescaler
10schlow enable
Internal clock fosc/4
Enable timer 1

$0,5 \mu s * (1:2) * 50000$
 $1 \mu s * 50000 = 50 \cdot 10^3 = 50 \text{ ms}$

0	0	1	1	1	1	0	0	1	0	1	1	0	0	0	0
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 = TMR1 = 15536
TMR1H TMR1L

tempo de incremento = $0,5 \mu s$

Prescaler = 1:2

TMR1H = 00111100, TMR1L = 10110000, TMR1 = 15536

CONFIGURAÇÃO T1CON:

-	-	0	1	1	1	0	1
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QUESTÃO 3 - timer2

$F_{osc} = 16 \text{ MHz}$ $T_{cm} = CM = \frac{4}{16 \text{ M}} = 0,25 \mu s$ $T_{int} = \frac{1}{1 \text{ kHz}} = 1 \text{ ms}$

$T_{int} = T_{cm} * \text{Prescaler} * (PR2 + 1) * \text{Postcaler} \rightarrow 1 \text{ ms} = 0,25 \mu s * \text{Prescaler} * (PR2 + 1) * \text{Postcaler}$

$4000 = \text{Prescaler} * (PR2 + 1) * \text{Postcaler} \rightarrow 4000 = 16 * (249 + 1) * 1$

$T_{cm} = 0,25 \mu s$

$1 \text{ ms} = 0,25 \mu s * \text{Prescaler} * (PR2 + 1) * \text{Postcaler}$

PR2 = 249

Prescaler 1:16, Postcaler 1:1

T2CON:

-	0	0	0	0	1	1	1
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